



1. AI-Powered Urban Traffic Near-Miss Detection

Problem Statement:

Urban intersections experience frequent accidents due to risky driving behaviour, sudden lane changes, red-light violations, and conflicts between vehicles and pedestrians. Conventional traffic monitoring systems primarily record violations after they occur and provide limited capability for identifying dangerous patterns before an accident happens.

Objectives:

- Develop a computer-vision system to detect vehicles, pedestrians, and traffic movements from video.
- Identify near-miss events such as sudden braking, unsafe overtaking, lane conflicts, and pedestrian-vehicle conflicts.
- Generate risk scores for road segments and intersections.
- Provide a dashboard showing high-risk locations and recurring patterns.

Expected Outcome:

- AI-based traffic risk analysis prototype.
- Automatic identification and classification of near-miss events.
- Interactive risk heatmap/dashboard for traffic authorities.
- Data-driven recommendations for improving road safety.

2. AI-Powered Waste Segregation Intelligence

Problem Statement:

Improper segregation of municipal waste reduces recycling efficiency and increases the amount of waste reaching landfills. Manual segregation is slow, labour-intensive, and inconsistent, while existing automated systems are often expensive.

Objectives:

- Develop a computer-vision model to classify common waste categories.
- Detect recyclable, organic, hazardous, and general waste.
- Provide disposal recommendations based on detected waste.
- Build a monitoring dashboard showing waste composition and segregation trends.

Expected Outcome:

- Working image-based waste classification system.
 - Automated waste-category prediction.
 - Disposal/recycling recommendation engine.
 - Analytics dashboard for monitoring waste patterns.
-

3. AI-Based Pothole and Road Damage Intelligence**Problem Statement:**

Road damage such as potholes, cracks, and uneven surfaces creates safety risks and increases vehicle maintenance costs. Reporting is often manual, resulting in delayed identification and repair prioritization.

Objectives:

- Detect potholes and road defects from images or video.
- Estimate severity and approximate location of detected damage.
- Prioritize road segments based on severity and frequency.
- Provide an interactive map for authorities to monitor reported damage.

Expected Outcome:

- Computer-vision road-damage detection prototype.
 - Automatic severity classification.
 - Location-based road condition map.
 - Repair-priority recommendation system.
-

4. Intelligent Public Transport Demand Predictor**Problem Statement:**

Public transportation systems frequently experience overcrowding during peak hours while operating underutilized vehicles during low-demand periods. Static schedules cannot effectively adapt to changing passenger demand.

Objectives:

- Analyse historical ridership, time, weather, holidays, and route data.
- Predict passenger demand for different routes and time periods.
- Identify expected overcrowding and underutilization.
- Recommend dynamic fleet allocation or scheduling adjustments.

Expected Outcome:

- Demand forecasting model.
 - Route-wise passenger prediction dashboard.
 - Overcrowding alerts.
 - AI-generated scheduling recommendations.
-

5. AI-Powered Campus Emergency Response Platform**Problem Statement:**

During campus emergencies, information is often fragmented across calls, messages, security personnel, and different departments. This can delay response and make it difficult for authorities to determine which incidents require immediate attention.

Objectives:

- Build a centralized platform for reporting campus emergencies.
- Automatically classify incidents based on severity and type.
- Prioritize incidents using AI-based risk scoring.
- Recommend the nearest available response team or resource.

Expected Outcome:

- Real-time emergency management dashboard.
 - AI-based incident classification and prioritization.
 - Response-team assignment system.
 - Incident tracking from reporting to resolution.
-

6. Intelligent Document Verification and Fraud Detection**Problem Statement:**

Organizations process large volumes of certificates, invoices, applications, and identity-related documents. Manual verification is time-consuming and vulnerable to manipulation, duplicated submissions, and inconsistent information.

Objectives:

- Extract structured information from uploaded documents using OCR.
- Detect inconsistencies between document fields.
- Identify suspicious alterations or duplicate submissions.
- Provide a confidence/risk score for each document.

Expected Outcome:

- AI-powered document analysis platform.
 - OCR-based structured information extraction.
 - Automated anomaly and duplication detection.
 - Verification dashboard with risk scores.
-

7. AI-Powered Food Freshness Assessment**Problem Statement:**

Food wastage occurs because consumers, retailers, and food-service operators often lack reliable information about the freshness of perishable products. Visual inspection alone can be subjective and may fail to identify early deterioration.

Objectives:

- Develop an AI model to analyse images of selected perishable food items.
- Classify food into freshness stages.
- Estimate remaining usable time using image and contextual data.
- Provide alerts for products approaching spoilage.

Expected Outcome:

- Computer-vision freshness assessment prototype.
 - Freshness-stage classification.
 - Remaining-life estimation interface.
 - Waste-reduction recommendation dashboard.
-

8. AI-Based Learning Gap Detection System**Problem Statement:**

Students within the same classroom can have significantly different levels of understanding. Conventional assessments often identify poor performance only after an examination, making early intervention difficult.

Objectives:

- Analyse quiz, assignment, and learning-performance data.
- Identify topic-wise knowledge gaps.
- Predict students at risk of falling behind.
- Recommend personalized learning resources and practice questions.

Expected Outcome:

- AI-powered student learning analytics platform.
 - Topic-level weakness detection.
 - Early-risk prediction.
 - Personalized learning recommendations.
-

9. Intelligent Queue and Crowd Prediction System**Problem Statement:**

Hospitals, railway stations, cafeterias, event venues, and public offices experience unpredictable crowding. Existing queue systems generally show current waiting conditions but provide limited ability to predict future congestion.

Objectives:

- Analyse historical and real-time queue/crowd data.
- Predict queue length and waiting time.
- Detect abnormal crowd growth.
- Recommend counter/resource allocation based on predicted demand.

Expected Outcome:

- Crowd and queue prediction engine.
 - Estimated waiting-time dashboard.
 - Congestion alerts.
 - Intelligent resource-allocation recommendations.
-

10. AI-Powered Local Disaster Risk Mapping**Problem Statement:**

Disaster management authorities often have access to large amounts of weather, geographic, infrastructure, and historical incident data, but these datasets are not always converted into actionable local risk information.

Objectives:

- Combine historical disaster, rainfall, terrain, infrastructure, and weather data.
- Generate localized risk scores.
- Identify areas vulnerable to flooding or other selected hazards.
- Display risk levels through an interactive map.

Expected Outcome:

- AI-based disaster-risk prediction prototype.
 - Localized risk scoring.
 - Interactive hazard map.
 - Early-warning and preparedness recommendations.
-

11. Smart Water Quality Monitoring and Contamination Prediction**Problem Statement:**

Small water bodies, community tanks, and groundwater sources may experience changing water quality due to contamination, seasonal variations, and surrounding activities. Periodic manual testing provides only occasional snapshots and may fail to identify deteriorating conditions early.

Objectives:

- Develop an IoT system for monitoring parameters such as pH, turbidity, TDS, and temperature.
- Build an AI model to identify abnormal water-quality patterns.
- Predict potential deterioration based on historical sensor readings.
- Generate alerts and recommended actions for users.

Expected Outcome:

- IoT water-monitoring prototype.
 - AI-based anomaly and trend prediction.
 - Real-time monitoring dashboard.
 - Automated water-quality alerts.
-

12. Smart Agriculture Irrigation Intelligence**Problem Statement:**

Farmers frequently irrigate crops using fixed schedules rather than actual soil and environmental conditions. This can lead to water wastage, over-irrigation, and reduced crop productivity.

Objectives:

- Monitor soil moisture, temperature, humidity, and other relevant parameters.
- Predict irrigation requirements using AI.
- Automatically recommend or trigger irrigation based on predicted need.
- Provide crop-zone-wise analytics.

Expected Outcome:

- Sensor-based smart irrigation prototype.
 - AI irrigation prediction model.
 - Automated irrigation decision system.
 - Farmer-oriented monitoring dashboard.
-

13. Cold-Chain Failure Prediction System**Problem Statement:**

Temperature-sensitive products such as food, vaccines, and certain pharmaceutical products can become unusable when exposed to unsuitable conditions during transportation and storage. Conventional monitoring often detects temperature violations only after they occur.

Objectives:

- Monitor temperature, humidity, and location throughout transportation.
- Detect abnormal environmental patterns.
- Predict potential cold-chain failures before critical thresholds are reached.
- Generate alerts and identify the likely stage of failure.

Expected Outcome:

- IoT cold-chain monitoring prototype.
 - AI-based failure prediction.
 - Real-time route/environment dashboard.
 - Early-warning system for temperature-sensitive goods.
-

14. Smart Building Energy Optimization**Problem Statement:**

Buildings consume significant energy because lighting, cooling, and electrical equipment are often operated without considering real-time occupancy and environmental conditions.

Objectives:

- Monitor electricity consumption, occupancy, temperature, and other relevant parameters.
- Predict energy demand.
- Identify abnormal or wasteful consumption.
- Recommend or simulate intelligent control strategies.

Expected Outcome:

- IoT-based energy monitoring system.
 - AI energy-demand prediction.
 - Appliance/zone-level consumption analytics.
 - Energy-saving recommendation engine.
-

15. Intelligent Solar Panel Fault Detection

Problem Statement:

Solar installations can experience reduced output due to dust, shading, overheating, damaged panels, wiring problems, and other faults. Manual inspection can be inefficient for large installations.

Objectives:

- Monitor panel voltage, current, temperature, and power output.
- Detect abnormal performance patterns.
- Classify possible fault conditions using AI.
- Estimate expected versus actual energy generation.

Expected Outcome:

- Solar monitoring prototype.
 - AI-based fault/anomaly detection.
 - Panel performance dashboard.
 - Maintenance-priority recommendations.
-

16. Smart Waste Bin Fill-Level Prediction

Problem Statement:

Fixed waste-collection schedules can result in overflowing bins in high-demand locations while collection vehicles visit partially empty bins unnecessarily.

Objectives:

- Monitor bin fill level using sensors.
- Analyse historical filling patterns.
- Predict when individual bins will reach capacity.
- Generate optimized collection recommendations.

Expected Outcome:

- Smart-bin monitoring prototype.

- Fill-level prediction model.
 - Overflow alerts.
 - Optimized collection-route recommendation system.
-

17. Intelligent Air Quality Monitoring Network

Problem Statement:

Air pollution can vary considerably across different locations and times of day. A limited number of monitoring stations cannot always provide sufficiently granular information for local communities.

Objectives:

- Build a distributed IoT monitoring system for air-quality parameters.
- Detect abnormal pollution patterns.
- Predict short-term air-quality changes.
- Generate location-specific alerts.

Expected Outcome:

- Distributed air-quality monitoring prototype.
 - AI-based pollution prediction.
 - Real-time pollution map.
 - Localized health/safety alerts.
-

18. Smart Parking Occupancy and Prediction System

Problem Statement:

Drivers waste time and fuel searching for parking spaces, particularly around campuses, commercial areas, hospitals, and event venues. Static parking information cannot predict availability during future peak periods.

Objectives:

- Detect parking-slot occupancy using sensors or computer vision.
- Predict parking availability based on historical patterns.
- Provide real-time slot information.
- Recommend suitable parking locations.

Expected Outcome:

- Smart parking monitoring prototype.
- Occupancy prediction model.

- Real-time parking dashboard.
 - Availability prediction and recommendation system.
-

19. Disaster-Resilient Emergency Communication Network

Problem Statement:

Floods, cyclones, earthquakes, and other disasters can disrupt conventional communication infrastructure, making coordination between rescue teams difficult. A temporary communication system is required to maintain connectivity even when centralized infrastructure becomes unavailable.

Objectives:

- Develop a prototype mesh/ad-hoc communication network.
- Monitor node connectivity and network health.
- Implement automatic rerouting when nodes become unavailable.
- Provide a dashboard for emergency communication and network monitoring.

Expected Outcome:

- Working resilient communication prototype.
- Self-healing network demonstration.
- Network-health monitoring dashboard.
- Priority communication mechanism for emergency teams.

This directly follows the resilience and self-healing communication direction present in the reference material.

20. Smart Worker Safety Monitoring System

Problem Statement:

Workers in construction, manufacturing, mining, and other hazardous environments may be exposed to excessive heat, toxic gases, abnormal motion, or accidents. Conventional safety equipment provides limited real-time awareness of these risks.

Objectives:

- Monitor environmental parameters such as temperature and gas concentration.
- Monitor worker motion and selected physiological parameters where feasible.
- Detect abnormal or dangerous conditions.
- Send immediate alerts to supervisors.

Expected Outcome:

- IoT-enabled worker safety prototype.
- AI-based abnormal-condition detection.
- Real-time supervisor dashboard.
- Emergency alert/SOS mechanism.

The concept is adapted from the embedded smart-helmet direction in your reference material.

21. Embedded Smart Helmet for Industrial Safety

Problem Statement:

Workers operating in construction and industrial environments face risks from toxic gases, excessive temperature, impacts, falls, and prolonged exposure to unsafe conditions. Conventional helmets provide physical protection but do not actively monitor the surrounding environment or worker condition.

Objectives:

- Design an embedded helmet prototype with environmental sensors.
- Detect gas, temperature, impact, and abnormal motion conditions.
- Include an emergency SOS mechanism.
- Establish wireless communication with a monitoring interface.

Expected Outcome:

- Functional embedded smart-helmet prototype.
- Real-time hazard detection.
- Emergency alert mechanism.
- Supervisor monitoring interface.

This retains the core hardware challenge from the supplied reference while making the deliverable appropriate for a hackathon prototype.

22. Smart Assistive Device for Visually Impaired Users

Problem Statement:

Visually impaired individuals may face difficulties identifying obstacles, navigating unfamiliar environments, and detecting potentially dangerous objects. Existing basic assistive devices often provide limited environmental awareness.

Objectives:

- Build a wearable or handheld embedded device.
- Detect nearby obstacles using suitable sensors.

- Provide directional or distance-based feedback.
- Integrate optional AI-based object recognition.

Expected Outcome:

- Functional assistive-device prototype.
 - Real-time obstacle detection.
 - Audio/vibration-based feedback.
 - Optional AI-powered object identification.
-

23. Intelligent Fire and Hazard Detection Node

Problem Statement:

Fire incidents can spread rapidly when early warning systems fail to detect abnormal temperature, smoke, or environmental conditions. Large buildings and industrial environments require distributed monitoring rather than relying on a single detection point.

Objectives:

- Design a compact embedded sensing node.
- Monitor temperature, smoke/gas, and other relevant parameters.
- Detect abnormal combinations of sensor readings.
- Transmit alerts to a centralized monitoring system.

Expected Outcome:

- Working multi-sensor fire-detection prototype.
 - Embedded anomaly-detection logic.
 - Wireless alert system.
 - Central monitoring dashboard.
-

24. Smart Energy Meter with Appliance-Level Intelligence

Problem Statement:

Conventional energy meters provide aggregate electricity consumption but do not explain which appliances are responsible for excessive usage or identify abnormal consumption patterns.

Objectives:

- Develop an embedded energy-monitoring device.
- Measure voltage, current, power, and energy consumption.
- Identify abnormal consumption patterns.

- Provide appliance-level or usage-category insights where feasible.

Expected Outcome:

- Functional smart-energy monitoring prototype.
- Real-time consumption measurement.
- Embedded anomaly detection.
- Energy analytics dashboard.



25. Flash Flood Detection & Early Warning System — Nepal

Problem Statement:

Nepal's mountainous terrain, intense monsoon rainfall, steep river catchments, and rapidly rising water levels can cause flash floods with very little warning. In vulnerable communities near rivers and downstream settlements, rainfall intensity can increase rapidly and overwhelm river or drainage capacity. Water level and its rate of rise can provide important indicators of immediate local flood risk, while rainfall alone may not capture the developing situation.

Conventional threshold-based warning systems can generate false alarms when individual measurements cross fixed limits, or miss dangerous situations when multiple moderate signals occur together. A data-driven system is therefore required to combine changing rainfall and water-level conditions and provide an early indication of short-horizon flood risk.

Objectives:

- Estimate short-horizon flood risk using rainfall intensity, water level, and rate-of-rise data trained on historical or realistically simulated flood scenarios.
- Develop a software-based inference engine capable of processing incoming or simulated sensor data in real time.
- Detect rapidly changing conditions that may indicate an approaching flash-flood event.
- Reduce false alarms by combining multiple environmental signals rather than relying only on fixed thresholds.
- Validate the system using simulated Nepal monsoon flood scenarios and evaluate detection accuracy, warning lead time, and false-alarm rate.
- Generate calibrated software-based alerts through a dashboard or application for communities and emergency-response authorities.

Expected Outcome:

- A trained ML-based flood-risk prediction model integrated into a real-time inference application.
- A simulation pipeline representing rainfall, water-level, and rate-of-rise conditions during Nepal-like monsoon flood scenarios.
- A real-time dashboard displaying sensor conditions, predicted flood risk, and warning status.
- An alert mechanism demonstrating the complete pipeline: data → ML prediction → risk decision → notification.
- Performance evaluation based on prediction accuracy, warning lead time, and false-alarm rate.